SHAHNAWAS ADEEL

Full-Stack Software Engineer | TypeScript, React, Node.js, PostgreSQL, Python

Chattogram, Bangladesh | +880 1533 970377 | shahnawasadeel@gmail.com | mdadeel.me | github.com/mdadeel | linkedin.com/in/shahnawasadeel

PROFILE

Full-stack engineer who takes products from schema to shipped interface. Built and deployed Arbor, a codebase audit platform that parses real repositories; easyui, a 36-component React design system with a design token pipeline; and Preen, an MIT-licensed Linux desktop utility with six tagged releases. Works end to end in TypeScript and Python, including background job processing, packaging and release tooling.

SELECTED PROJECTS

Arbor

Live site / Code

Codebase audit platform for GitHub repositories. Parses a connected repo, draws its dependency graph, and scores six health dimensions with file and line evidence attached to every finding.

- Wrote the analysis layer in strict TypeScript: Babel AST parsing of JavaScript and TypeScript sources, import resolution into a React Flow module graph, cycle and hotspot detection, secret and env checks, and a schema-validated SARIF 2.1.0 export that uploads into GitHub Code Scanning.
- Modeled 30 Prisma models on PostgreSQL 16 across 11 migrations and exposed 19 tRPC v11 routers for projects, workspaces, billing, sharing and admin.
- Ran scans as BullMQ jobs on Redis 7: each job shallow clones into an isolated temp directory, streams progress, recovers stale jobs on worker restart, and deletes the clone when it finishes. Cache layer keeps scans under the per-workspace quota.
- Packaged self-hosting as one **docker compose up**: bundled PostgreSQL and Redis, migration runner, health-checked Next.js app and a long-lived worker container. Added headless Chromium A4 audit PDFs and AES-256-GCM encryption for stored provider keys.

Stack: Next.js 15 App Router, React, tRPC, Prisma, PostgreSQL, Redis, BullMQ, Docker, Playwright, Vitest (68 test files). AGPL-3.0.

easyui

Live site / Code

React component library and design token pipeline. Two published workspace packages and a live gallery of 44 examples, all themeable at runtime.

- Split the system into **@easyui/react** (36 accessible primitives, each with its own test file) and **@easyui/tokens** (JSON design tokens compiled to CSS custom properties at build time), managed as a pnpm workspace monorepo.
- Built runtime theming through EasyUIProvider: accent, radius and font change over CSS variables with no rebuild, and preferences persist per device.
- Covered keyboard interaction, focus management and ARIA states with 44 Vitest and axe accessibility test files, then deployed the playground to Cloudflare Workers with prerendered routes, per-page metadata, structured data and a generated sitemap so pages stay crawlable without JavaScript.

Preen

Code / Releases

Linux system utility in Python 3 and GTK 3. Manages APT, Flatpak and AppImage packages, previews cleanup sizes, inspects systemd services and archives user configuration.

- Maintain an 8.6k-line desktop application for Debian and Ubuntu family systems, including a simulate mode that previews destructive commands, a blocked list of essential packages on uninstall paths, and polkit authorization for privileged actions.
- Automated .deb, AppImage and wheel builds through shell packaging scripts, cut six tagged releases up to v0.7.1, and kept 17 test modules with fakes for apt, flatpak and systemd so the suite runs without root.
- Wrote the security, privacy, user and architecture documentation a distro package needs. MIT licensed, with an external fork in use.

e-TuitionBD

Live site / Code

Multi-role tuition marketplace for Bangladesh. Student, tutor, organization and admin dashboards on one React codebase.

- Built role-scoped dashboards and application flows in React 19 and Vite, with Firebase Auth, Stripe checkout, Socket.IO messaging, Bangla and English locales, and Playwright end-to-end tests for payment approval and tutor onboarding.

EXPERIENCE

Freelance Software Engineer | Client and product work — Remote | 2025 - Present

- Shipped **Metro Optics**, an eyewear storefront in React and Tailwind CSS, covering catalog filtering, product detail, cart and checkout screens across mobile and desktop breakpoints.
- Delivered the Ayash Tech corporate site in React with React Hook Form validation, deployed and tuned on Vercel.
- Own delivery end to end: scoping with the client, component architecture, responsive QA and post-launch fixes.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript (ES2024), Python 3, SQL, Bash
Frontend	React 19, Next.js (App Router), Vite, Tailwind CSS, Radix UI, TanStack Query, React Flow, Recharts
Backend	Node.js, tRPC v11, Express, BullMQ, Socket.IO, NextAuth, Firebase Auth, Stripe, REST APIs
Data	PostgreSQL 16, Prisma, MongoDB, Redis 7 (ioredis), schema design and migrations
Tooling	Docker and Compose, Playwright, Vitest, pnpm workspaces, GitHub Actions, Vercel, Cloudflare Workers, Git
Systems	Python 3, GTK 3, PyGObject, Debian packaging, AppImage, systemd, apt and flatpak
Spoken	Bangla (native), English (professional), Hindi (conversational)

EDUCATION AND TRAINING

MERN Stack Web Development | Programming Hero — 2025 - Present

Coursework in React, Node.js, Express and MongoDB; 500+ JavaScript, data structure and problem-solving exercises.